

Birinder Bachhal

bachhalbirinder1@gmail.com | (516) 800-1841 | U.S. Citizen
linkedin.com/in/birinder-bachhal | bbachhal.github.io

Education

Boston University

Kilachand Honors College / College of Engineering

Degree: B.S. Mechanical Engineering | Concentrations: Robotics, Manufacturing

Awards/Organizations: QuestBridge Scholar, F1Tenth ACRO, Theta Tau Professional Engineering Fraternity, Society of Asian Scientists and Engineers, Sikh Association at BU

Boston, MA

September 2024 – May 2028

Technical Skills

CAD/CAE: SolidWorks, Fusion 360, AutoCAD, GD&T, Tolerance Stack-Up Analysis, DFM/DFA, FEA, Thermal Analysis, CAM, Technical Drawings

Robotics: Universal Robots (UR12e), RoboDK, Robot Cell Design, End-Effector Design, Offline Programming, Path Planning, LIDAR/Sensor Integration

Manufacturing: 5-Axis CNC Machining, Fixture & Workholding Design, Silicone Molding & Demolding, FDM/SLA 3D Printing, Rapid Prototyping, Additive Manufacturing

Test & Quality: Six Sigma, Metrology, Dimensional Inspection, First-Article Inspection, Root Cause & Failure Analysis, Design Verification, Process Capability

Controls & Data: Rockwell PLCs, Relay Logic, Machine Safety Systems (Light Curtains, E-Stop), Arduino, Raspberry Pi, PID Control, Stepper Motor Control, Pneumatics, Python, MATLAB, Power BI

Work Experience

The Renatural

Engineering Intern, Robotics & Manufacturing

Brooklyn, NY

July 2026 – August 2026

- Built a Universal Robots UR12e robotic work cell from scratch, cutting production cycle time from 90 to 45–60 minutes; designed 22 custom components across a 50-piece structural frame integrating Keyence light curtains and E-stop safety interlocks
- Engineered end-effector mounts for a Zimmer electric gripper and proprietary injection head; validated toolpaths in RoboDK simulation prior to hardware deployment, executing 50–100 precision injections per part
- Designed 3 custom 5-axis CNC workholding fixtures and a pneumatic silicone demolding vessel, applying tolerance stack-up analysis to eliminate part shift; raised usable yield 20% and removed operator entry from the envelope
- Characterized an uncharacterized natural fiber to establish process capability where no material datasheet existed; troubleshot Rockwell PLC and relay module faults and presented process findings to company leadership

Deteqt

Mechanical Engineering Intern

Sydney, Australia

January 2026 – June 2026

- Designed IP65-rated ruggedized enclosure in SolidWorks for 24-hour field deployment above 40°C, delivering GD&T drawings across 4 FDM and SLA prototype iterations verified by dimensional inspection
- Engineered aluminum baseplate heatsink dissipating 2–5W thermal load while preserving quantum sensor performance, applying thermal modeling and DFM for CNC production
- Released a complete production documentation package covering fastener standardization, heat-set insert specifications, and assembly drawings; coordinated design transfer to an external supplier that built to print with zero engineering follow-up

Projects

BU F1Tenth Autonomous Racing

Vice President (Hardware)

Boston, MA

December 2024 – Present

- Leading mechanical design on a Guinness World Record attempt converting a 2008 Corvette to an autonomous platform
- Redesigned 1:10 scale chassis in SolidWorks, improving structural rigidity 30% and reducing weight 15% through FEA-driven iteration; developed bench test procedures verifying LIDAR and servo communication before track deployment

The Deskinator – Autonomous Workspace Robot

Design Lead

Boston, MA

September 2025 – December 2025

- Designed and machined custom housing integrating NEMA-17 stepper motors and proximity sensors; developed PID control logic on Raspberry Pi across 3 prototype iterations
- Performed root cause analysis on first-article failures, redesigning motor mounts and sensor fixtures to eliminate recurring edge-detection faults

OrthoCAD Pro – Automated Orthotic Design Platform

Personal-Interest Project

Boston, MA

July 2025 – September 2025

- Built a smartphone photogrammetry pipeline achieving ± 1 mm scan repeatability, automating manufacturing-ready parametric CAD generation in Fusion 360

First-Year Innovation Fellowship – Closed-Loop Insulin Delivery

2024-2025 Fellow

Boston, MA

September 2024 – June 2025

- Designed 2 functional prototypes integrating mechanical hardware with embedded control logic modeling real-time glucose regulation

Leadership

Vice President, BU F1Tenth Autonomous Racing | **Executive Board**, Sikh Association at BU | **Academic Chair**, Theta Tau